Project 1 : Rock, Paper, Scissors Game :

→ We all have played Rock, paper Scissors game in our childhood.
Write a python program capable of playing this game with user.

Python Code :

```
Import random
Computer = random.choice ([1,2,3])

you Str= input ("Enter your choice : "). lower ()
you Dict = ({"rock": 1, "paper": 2, "scissors": 3}).

rev Dict = { 1 : "rock", 2 : "paper", 3 : "scissors"}
reverse = revDict [Computer]

You = you Dict [you Str]

Print ("Computer choice : ", reverse)

if you == Computer :
    print ("Its a tie")
elif you == 1 and Computer == 3 or you == 2 and
    Computer == 1 or you == 3 and Computer == 2 :
    print ("You win !")

else :
    print ("Computer wins")
```

1. First : Importing Random Module.

```
Import random
```

- This allows the computer to make random choices.
- Without this, the computer cannot randomly select rock / paper / scissors.

2 → Computers Random Choice

```
Computer = random.Choice([1, 2, 3])
```

- This line of code randomly selects one number.
- 1 → rock
- 2 → Paper
- 3 → Scissors.

★ The selected number is stored in variable Computer.

So now the computer has secerity choosen something.

3 → Taking Users Input.

```
Youstr = input("Enter your choice").Lower()
```

- Input takes the input from the user.
- .Lower() converts input to lowercase

4. Dictionary for User Input :

```
YouDict = {"rock" : 1, "Paper" : 2, "Scissors" : 3}
```

- This dictionary converts words into numbers.

5. Reverse Dictionary for Display :

```
revDict = {1: "rock", 2: "paper", 3: "scissors"}
```

- This does the opposite.
- It converts numbers back to words.

Why ?
- Because Computer choose the number.
- But we want to display it as a word

6. Getting Computer's Choice in Words :

```
reverse = revDict [computer]
```

- Computer contains 1, 2, 3.
- We use revDict to convert it into text.

e.g. if computer = 2
reverse = "paper"

7. Converting User Choice to Number

```
you = youDict [youStr]
```

- if user typed 'rock'
- you = 1

Now Both user and Computer are numbers.

8. Game logic :

```
if you == Computer:
    print ("It's a tie")
```

If both choose same number → Tie

```
elif you = 1 and Computer == 3 or you == 2 and
    computer == 1 or you == 3 and computer == 2:
        print ("you win")
```

- This checks winning conditions:

```
else:
    print ("Computer Win")
```

- If It's not tie and not you win then computer wins.